



Clinical Practice Guideline

Periprocedural Management of the Anticoagulants

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Home Anticoagulant Support Service (HAAS), Haematology	Medical

Introduction

The perioperative management of anticoagulation needs to consider the thromboembolic risk associated with stopping anticoagulation, and the risk of bleeding associated with the procedure and the patient's individual characteristics. Assessment of the patient should include determining these risks, as well as determining the time of anticoagulation interruption and whether bridging anticoagulation is required.

This document should be utilised in conjunction with SCGOPHG Medicines Management Policies and Procedures

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☐	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☐
Other:	☐		

Guiding Principles

General approach to periprocedural anticoagulation management

Both patient-related and surgical risk factors for thrombosis and bleeding should be assessed to determine an overall peri-procedural anticoagulation management plan for each individual patient. This often requires input from multiple teams (including cardiology, haematology, surgical and anaesthesia teams).

An overall approach to management is as follows and will be detailed in this guideline:

1. Estimate thromboembolic risk
2. Estimate patient bleeding risk
3. Estimate the procedural bleeding risk.
4. Management of the anticoagulant:
 - a. Determine the timing of anticoagulation interruption.
 - b. Determine whether bridging anticoagulation is necessary.

Occasionally, in patients who require elective surgery, it may be safer to postpone the procedure. Examples of such situations include:

- Post stent placement.
- Within three months following a VTE, arterial thromboembolism or stroke.
- Recent prosthetic valve insertion.

In these circumstances, it is best to speak to the patient's specialist to decide on the optimal timing of the procedure.

Bridging anticoagulation

Bridging anticoagulation refers to the administration of a short-acting anticoagulant to cover the period off regular anticoagulation perioperatively. The intention of bridging is to minimise the time the patient is not anticoagulated, thereby minimising the risk of thromboembolism. This needs to be balanced against the risk of bleeding, which is increased when bridging anticoagulation is used.^{1,7} Emerging evidence from large meta-analyses has shown that bridging does not significantly reduce thromboembolic risk, at the expense of increased bleeding risk. Therefore, there are a limited number of indications for bridging anticoagulation. In most cases, these patients are on warfarin.

1. Estimating the periprocedural thrombotic risk

Common reasons for patients to be anticoagulated include Atrial Fibrillation (AF), venous thromboembolism (VTE) or mechanical heart valves. Estimation of a patient's risk of thrombosis includes their indication for anticoagulation, co-morbidities and age.

Table 1: Risk stratification for perioperative thrombotic risk^{10,14}

Risk category	Mechanical heart valve	Non valvular AF	VTE
High: > 10%/yr risk of ATE, >10% /mn risk of VTE	• Any mitral valve prosthesis • Caged ball or tilting disk valve in mitral/aortic position • Recent (< 6 months) stroke or TIA	• Recent (< 3 mn) stroke or TIA • Rheumatic heart disease • CHA₂D₂-VASc Score 7-9**	• Recent (< 3 mn) VTE • Severe thrombophilia (deficiency of protein C, protein S, or ATIII, APL antibodies, multiple thrombophilias) • Thrombosis associated with IVC filter
Intermediate: 4–10%/yr risk of ATE or 4–10%/mn risk of VTE	• Bileaflet prosthetic aortic valve with major risk factors for stroke*	• CHA₂D₂-VASc Score 4-6	• VTE within 3-12 mn • Recurrent VTE • Non-severe thrombophilia (heterozygous factor V Leiden or prothrombin gene mutation) • Active cancer^
Low: < 4%/yr ATE or < 2%/mn VTE	• Bileaflet prosthetic aortic valve without major risk factors for stroke	• CHA₂D₂-VASc score 0-3	• VTE > 1 yr ago and no other risk factors

ATE = arterial thromboembolism, VTE = venous thromboembolism, TIA = transient ischaemic attack, mn=month, yr = year, ATIII = antithrombin III, APL= antiphospholipid, IVC = inferior vena cava

*Risk factors for stroke: AF, prior stroke or TIA, hypertension, diabetes, congestive heart failure, age >75 yr.

**The CHA₂DS₂VaSc score is a scoring system used to determine the annual risk of stroke in AF.

^ Certain cancers such as pancreatic cancer, have a higher thrombosis risk; this should be considered in each individual

Table 2: CHA₂DS₂VASC score

Risk factor	Score		CHA ₂ DS ₂ VaSc score	Annualised stroke risk (%/year)
Congestive heart failure	1		0	0%
Hypertension	1		1	1.3%
Age ≥ 75	2		2	2.2%
Age 65-74	1		3	3.2%
Diabetes	1		4	4.0%
Stroke/TIA/VTE	2		5	6.7%
Vascular disease	1		6	9.8%
Female	1		7	9.6%
			8	6.7%
			9	15.2%

2. Estimating the patient bleeding risk

In addition to the bleeding risk associated with the type of surgery, there are also patient-specific risk factors which can increase risk for bleeding further. Therefore, these risk factors also need to be taken into consideration when managing anticoagulation perioperatively¹⁰. Several risk scoring algorithms exist to predict patient's bleeding risk. The HEMORR₂HAGES score was designed to quantify bleeding risk in patients with Atrial Fibrillation over the age of 75 years of age.

Table 3: HEMORR₂HAGES score¹¹

	Clinical characteristics	Score
H	Hepatic or renal disease	1
E	Ethanol abuse	1
M	Malignancy	1
O	Older age ≥ 75	1
R	Reduced platelet count or function	1
R	Rebleeding	2
H	Hypertension (uncontrolled)	1
A	Anaemia	1
G	Genetic risk factors (CYP 2C9 snp)	1
E	Excess falls risk	1
S	Stroke	1

Score	Bleeds per 100 patient years of warfarin (%)
0	1.9
1	2.5
2	5.3
3	8.4
4	10.4
≥ 5	12.3

3. Estimate the procedural bleeding risk^{6,17}

The listed risk stratification is a guide only and is not complete. Seek specialist advice if uncertain as the proceduralist is best suited to risk stratification of the procedure.

Table 5: Suggested risk stratification for procedural bleeding risk⁶

Minimal bleeding risk procedure	Low to moderate bleeding risk procedure	High bleeding risk procedure
Consider continuing anticoagulation	2-day risk of major bleed <2%	2-day risk of major bleed ≥2%

Minor dermatologic procedures Cataract procedures Minor dental procedures Pacemaker or defibrillator device implantation Minor breast surgery Carpal tunnel surgery	Endoscopies ± biopsy Coronary angiography with femoral access Bronchoscopy ± biopsy Laparoscopic cholecystectomy Abdominal hernia repair Shoulder/foot/hand surgery Endovascular procedures Lymph node biopsy Haemorrhoidal surgery	Major surgery with extensive tissue injury Cancer surgery Major orthopaedic surgery Major thoracoabdominal surgery Urologic or gynaecologic surgery TURP, bladder resection tumour ablation Surgery or biopsy in highly vascular organs (kidneys, liver, spleen) Bowel resection Cardiac, intracranial or spinal surgery Thyroid surgery Vascular surgery Posterior eye chamber procedures Deep nerve block Certain gastrointestinal procedures (e.g. ERCP)
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4. Management of the anticoagulants

Consider the thrombotic risk and procedural bleeding risk as above.

Periprocedural management of the DOACs^{5,6,17,18}

The periprocedural management of the DOACs depends on the pharmacokinetics of the drugs and the patient's renal function.

In most cases, there is no indication for perioperative bridging when patients are on DOACs³. Seek advice if the patient's procedure is delayed and anticoagulation has been ceased.

If a patient requires urgent surgery and has had a CVA, VTE or stent placement in the last 6 months, seek guidance from the relevant specialist for advice on appropriate management.*

**Neurologist, haematologist, cardiologist or vascular surgeon*

Timing of DOAC cessation pre-operatively

Drug	Renal function	Low bleeding risk	High bleeding risk
Apixaban 5mg bd	CrCl >30 ml/min	Omit 1 day before	Omit 2 days before
Apixaban 2.5mg bd	CrCl ≥ 25 ml/min	Omit 1 day before	Omit 2 days before
Rivaroxaban 20mg od	CrCl > 30ml/min	Omit 1 day before	Omit 2 days before
Rivaroxaban 10mg od	CrCl ≥ 15 ml/min	Omit 1 day before	Omit 2 days before
Rivaroxaban 2.5mg bd	CrCl ≥ 15 ml/min	Omit 1 day before	Omit 2 days before
Dabigatran	CrCl ≥ 50 ml/min	Omit 1 day before	Omit 2 days before
Dabigatran	CrCl <50ml/min	Omit 2 days before	Omit 4 days before

For all the anticoagulants, omit the anticoagulant on the day of surgery

Recommencing DOACs post-operatively

The managing surgeon or proceduralist will generally decide when to recommence anticoagulation. As a general guide:

Bleeding risk of procedure	Post-op anticoagulant management	
Low bleeding risk	Commence regular DOAC on day 1 post op	
High bleeding risk	Commence regular DOAC 48-72 hr post op.	If continued high bleeding risk, consider prophylactic dose DOAC (i.e apixaban 2.5mg bd or rivaroxaban 10mg od) until safe to resume therapeutic dosing.

Post-operative VTE prophylaxis may be considered in patients with a high risk of thromboembolism until resumption of the DOAC, when safe to do so, as per the local VTE prophylaxis

Periprocedural management of warfarin

Warfarin has a long half-life and the anticoagulant effect usually takes several days to wear off.

Bridging during warfarin interruption

In patients at high thrombotic risk (such as those with mechanical heart valves or AF with high CHA₂DS₂VASc scores), bridging with a short acting anticoagulant (usually LMWH), is used to reduce the time off anticoagulation and thereby minimise risk of thrombosis. In most patients, bridging is not required as the risk of bleeding is greater than the thrombotic risk^{1,6}.

Bridging may be appropriate during warfarin interruption in the following situations^{7,8,9}:

- Mechanical mitral valve or mechanical aortic valve with stroke risk factors
- Embolic stroke within the last three months or very high stroke risk (e.g. high CHA₂D₂-VASc score).
- Proximal VTE in the previous three months.
- Prior thromboembolism during previous anticoagulation interruption¹⁰.

If there is uncertainty regarding the need for bridging anticoagulation, recommend consultation with the specific specialty involved in the patient's care or referral to the Home Anticoagulation Service (HASS) for assessment.

Low bleeding risk procedure

Continue warfarin if INR is in range. Consider a lower perioperative INR goal of 2.0.

Intermediate or high bleeding risk procedure

Low thrombotic risk	Intermediate thrombotic risk	High thrombotic risk
Stop warfarin 4-5 days before surgery		
No bridging necessary	Consult haematology/ cardiology / HASS to discuss bridging options	
	If used, consider prophylactic dose LMWH 2-3 days before surgery if INR <2	If therapeutic dose LMWH is used, start 2-3 days before surgery when INR is sub-therapeutic. Last LMWH dose 24 hours before surgery ^(3, 4)
Restart warfarin on day of surgery if appropriate		
	Restart bridging anticoagulation once haemostasis is adequate. For patients at high bleeding risk, consider starting prophylactic dose LMWH for 2-3 days first	

LMWH = low molecular weight heparin

Periprocedural management of heparin⁶

Pre-operative cessation of unfractionated heparin and low molecular weight heparin is based on the half-lives of the anticoagulants.

Postoperative resumption of unfractionated heparin and low molecular weight heparin is similar, based on the onset of anticoagulation at approximately one hour after administration for both forms of heparin, and peak anticoagulant activity at approximately three to five hours.

Unfractionated heparin

The half-life of unfractionated heparin is 45 minutes.

Dosing of UFH	Pre-operative cessation	Post-operative resumption
Subcutaneous prophylactic dose	Last dose evening before	Recommence at next scheduled dose provided that haemostasis is normal and surgeon has approved it
Intravenous infusion	Stop 5-6 hours prior to the procedure	Minor procedures, low bleed risk: recommence therapeutic dose 24 hour post procedure
	<i>In patients with severe coronary disease and unstable angina going for a CABG, the infusion is continued until the time of surgery. Consult the treating specialist for confirmation</i>	Major surgery or high bleeding risk patients: recommence therapeutic dose 48-72 hours post procedure.

Low molecular weight heparin

The half-life of enoxaparin is 4.5 to 7 hours; the half-life of dalteparin is 3-5 hours¹⁶.

Dosing of enoxaparin	Pre-operative cessation	Post-operative resumption
Prophylactic dose (e.g. enoxaparin 20mg of 40mg od)	Withhold 12 hours prior to procedure	Recommence 4-6 hours post procedure, or longer if there were complications or the patient has a high bleeding risk
Therapeutic dose (e.g. enoxaparin 1mg/kg bd or 1.5mg/kg od)	Discontinue 24 hours prior to the procedure. The last dose should be given the morning BEFORE the procedure	Minor procedures, with low bleeding risk, therapeutic dose can usually recommence 24 hours post procedure.
		Major surgery or high risk of bleeding, recommence therapeutic dose 48-72 hours post procedure

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